Light

Network Hardware & Topologies

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OCR GCSE Computer Science — **Networks and Topologies (J277/01 Section 1.3)**. This topic covers types of networks (LAN and WAN), network hardware components, star and mesh topologies, the internet and DNS, and the differences between client-server and peer-to-peer network models.

Exam questions typically ask you to **describe** the purpose of network hardware, **compare** topologies or network models, and **explain** factors affecting network performance. Questions range from 1-mark definitions to 6-mark comparisons requiring analysis of advantages and disadvantages.

What you must know

This checklist covers the key exam-visible concepts. You should be able to explain each point and give an example where appropriate.

- **Define** LAN (Local Area Network) and WAN (Wide Area Network) and explain the differences between them.
- **Describe** the purpose of network hardware: NIC, switch, router, WAP, and transmission media.
- **Draw and describe** star topology and mesh topology, including advantages and disadvantages of each.

- **Explain** what the internet is and how it differs from a single network.
- **Describe** the role of DNS (Domain Name System) in converting domain names to IP addresses.
- **Explain** the concept of hosting and cloud computing.
- **Compare** client-server and peer-to-peer network models.
- **Describe** factors that affect network performance: bandwidth, latency, number of users, transmission media, interference.

> **Specification note.** OCR J277 Section 1.3 requires understanding of LANs, WANs, network hardware, star and mesh topologies, the internet (including DNS, hosting, and the cloud), and client-server versus peer-to-peer models. Questions frequently combine hardware identification with topology diagrams.

Toolkit

LAN vs WAN — key differences

Feature	LAN (Local Area Network)	WAN (Wide Area Network)
Geographic area	Small area (single building, campus)	Large area (cities, countries, worldwide)
Ownership	Usually owned by one organisation	Often uses third-party infrastructure
Speed	Higher speeds (typically 1 Gbps+)	Lower speeds (depends on connection)
Cost	Lower setup and maintenance costs	Higher costs (leased lines, equipment)
Examples	School network, home network, office	The internet, bank networks across branches

Network hardware — essential components

Hardware	Purpose	Key point
NIC (Network Interface Card)	Allows a device to connect to a network; has a unique MAC address	Every networked device needs one (often built-in)
Switch	Connects devices within a LAN; sends data only to the intended recipient using MAC addresses	More efficient than a hub (which broadcasts to all)
Router	Connects different networks together; directs data packets	Essential for connecting LANs to

Hardware	Purpose	Key point
	between networks using IP addresses	the internet
WAP (Wireless Access Point)	Allows wireless devices to connect to a wired network	Creates a Wi-Fi hotspot; uses radio waves
Transmission media	Physical means of transmitting data: copper cable, fibre optic, wireless	Fibre optic is fastest; wireless is most convenient

Star vs Mesh topology — comparison

Feature	Star topology	Mesh topology
Structure	All devices connect to a central switch/hub	Devices connect to multiple/all other devices
Single point of failure	Yes — if central switch fails, network fails	No — multiple paths available
Fault tolerance	One device failure doesn't affect others	Very high — data can take alternative routes
Cost	Lower — fewer cables needed	Higher — many cables and connections
Complexity	Simple to set up and manage	Complex to set up and maintain
Use cases	Most LANs (schools, offices, homes)	Critical systems, military, internet backbone

Client-server vs Peer-to-peer

Feature	Client-server	Peer-to-peer (P2P)
Central server	Yes — dedicated server(s) provide services	No — all computers are equal
Security	Centralised, easier to manage	Decentralised, harder to secure
Backups	Centralised on server	Each user responsible for own data

Feature	Client-server	Peer-to-peer (P2P)
Cost	Higher (server hardware, software, staff)	Lower (no dedicated server needed)
Scalability	Good — can add more clients easily	Poor — performance drops as network grows
Examples	School networks, company email, web servers	File sharing, small home networks, gaming

Exam tip: When comparing topologies or network models, always give at least one advantage and one disadvantage of each. A balanced comparison scores higher marks than a one-sided answer.

1. Network types and hardware

A **network** is two or more computers connected together to share resources and communicate. Networks allow users to share files, printers, and internet connections, and enable communication through email and messaging. Understanding the different types of networks and the hardware that makes them work is fundamental to computer science.

1.1 Local Area Network (LAN)

A **LAN (Local Area Network)** is a network that covers a small geographic area, such as a single building, school, or home. Key characteristics of a LAN include:

- **Limited geographic area:** Usually confined to one site or building
- **High speed:** Typically 100 Mbps to 10 Gbps or faster
- **Single ownership:** Usually owned and managed by one organisation
- **Private infrastructure:** Uses privately owned cables and equipment
- **Low latency:** Fast response times due to short distances

LANs are the most common type of network. Your school network is a LAN — all the computers, printers, and servers are connected within the school buildings using cables and wireless access points owned by the school.

A typical school LAN allows students to log in from any computer and access their files, use shared printers, access the internet through a central connection, and use educational software installed on servers.

1.2 Wide Area Network (WAN)

A **WAN (Wide Area Network)** is a network that covers a large geographic area, potentially spanning cities, countries, or even the entire world. The internet is the largest WAN in existence.

- **Large geographic area:** Can span cities, countries, or continents
- **Lower speeds:** Generally slower than LANs due to distance
- **Third-party infrastructure:** Often uses leased lines from telecommunications companies
- **Higher latency:** Greater delays due to longer distances
- **Higher cost:** Expensive to set up and maintain

A bank with branches across the country would use a WAN to connect all branches together, allowing customers to access their accounts from any location and enabling the bank to manage all its data centrally.

1.3 Network hardware

Several hardware components are needed to create and operate a network. Each has a specific purpose:

Network Interface Card (NIC): A NIC is a hardware component that allows a device to connect to a network. Every networked device needs a NIC — without one, the device cannot send or receive data on the network. Modern computers, laptops, and phones have NICs built in. Each NIC has a unique **MAC address** (Media Access Control address) — a permanent hardware identifier assigned during manufacture. MAC addresses are used by switches to direct data to the correct device within a LAN.

Switch: A switch connects multiple devices within a LAN. When data arrives at the switch, it reads the destination MAC address and forwards the data only to the port where that device is connected. This is more efficient than a hub, which would broadcast data to all connected devices. Switches operate at the data link layer and use MAC addresses to make forwarding decisions.

Router: A router connects different networks together — for example, connecting a LAN to the internet (which is a WAN). Routers use **IP addresses** to determine where to send data packets. When your computer requests a web page, the router directs that request out to the internet and then routes the response back to your specific device. Routers can also provide DHCP (assigning IP addresses) and NAT (allowing multiple devices to share one public IP address).

Wireless Access Point (WAP): A WAP allows wireless devices (laptops, phones, tablets) to connect to a wired network. The WAP acts as a bridge between the wireless devices and the wired network infrastructure. It transmits data using radio waves at specific frequencies (typically 2.4 GHz or 5 GHz). Most home "routers" are actually combination devices that include a router, switch, and WAP in one unit.

1.4 Transmission media

Transmission media refers to the physical materials used to transmit data:

- **Copper cable (Ethernet):** Most common in LANs; uses electrical signals; Cat5e/Cat6 cables; speeds up to 10 Gbps; susceptible to electromagnetic interference
- **Fibre optic cable:** Uses light pulses; very high speeds (100+ Gbps); not affected by interference; more expensive; fragile
- **Wireless (Wi-Fi):** Uses radio waves; convenient and flexible; slower than wired; can suffer from interference and range limitations

> **Exam tip:** Don't confuse a switch with a router. A switch connects devices within the same network (using MAC addresses). A router connects different networks together (using IP addresses).

Worked Example 1 – Defining LAN and WAN

Question: Define the terms LAN and WAN. [2 marks]

Model answer:

LAN: A Local Area Network is a network that covers a small geographic area such as a single building or site **(1)**.
WAN: A Wide Area Network is a network that covers a large geographic area, potentially spanning cities, countries, or worldwide **(1)**.

Exam tip: The key difference is geographic area. LAN = small/local, WAN = large/wide. Always mention the geographic scope in your definitions.

Worked Example 2 — Purpose of a switch

Question: Describe the purpose of a switch in a network. [2 marks]

Model answer:
A switch connects multiple devices within a LAN **(1)**. It uses MAC addresses to forward data only to the intended recipient device rather than broadcasting to all devices **(1)**.

Exam tip: Mention that switches use MAC addresses and only send data to the correct device. This distinguishes a switch from a hub.

Worked Example 3 — Router vs switch

Question: Explain the difference between a router and a switch. [2 marks]

Model answer:
A switch connects devices within the same network (LAN) using MAC addresses **(1)**. A router connects different networks together (e.g., LAN to internet) using IP addresses **(1)**.

Exam tip: The key distinction is: switch = within one network, router = between different networks. Mention the addressing type (MAC vs IP) for full marks.

Worked Example 4 — Network hardware identification

Question: A school wants to allow students to connect their phones to the school network wirelessly. State the hardware device needed and explain its purpose. [2 marks]

Model answer:

A Wireless Access Point (WAP) is needed **(1)**. It allows wireless devices to connect to the wired network by transmitting data using radio waves **(1)**.

Exam tip: WAP questions often appear in scenarios about adding wireless connectivity. Remember: WAP = bridge between wireless devices and wired network.

2. Topologies and the internet

Network **topology** refers to the arrangement of devices and connections in a network. The topology affects performance, reliability, and cost. You need to know two topologies: star and mesh. You also need to understand how the internet works, including DNS, hosting, and cloud computing.

2.1 Star topology

In a **star topology**, all devices connect to a central switch or hub. All data passes through this central device. This is the most common topology for LANs.

How it works: When Computer A wants to send data to Computer B, the data goes from A to the central switch, which then forwards it to B. No device communicates directly with another — everything goes through the centre.

Advantages of star topology:

- If one device or cable fails, only that device is affected — other devices continue working
- Easy to add new devices — just connect to the central switch
- Easy to identify and fix faults — problems are isolated to individual connections
- Good performance — dedicated connection to the switch for each device
- Relatively simple to set up and manage

Disadvantages of star topology:

- **Single point of failure:** If the central switch fails, the entire network goes down
- Requires more cable than a bus topology (each device needs its own cable to the switch)
- The central switch can become a bottleneck if traffic is very high
- Cost of the central switch (though switches are relatively inexpensive now)

2.2 Mesh topology

In a **mesh topology**, devices are connected to multiple other devices, creating multiple possible paths for data. In a **full mesh**, every device is connected to every other device. In a **partial mesh**, some devices have multiple connections but not to every other device.

How it works: When Computer A wants to send data to Computer B, the data can take multiple possible routes. If one path fails, data is automatically rerouted through another path.

Advantages of mesh topology:

- **No single point of failure:** Multiple paths mean the network continues working even if connections fail
- High fault tolerance and reliability — ideal for critical systems
- Data can take the fastest available route, improving performance
- Easy to identify faults — problems don't cascade through the network

Disadvantages of mesh topology:

- **High cost:** Requires many cables and network interfaces

- Complex to set up, configure, and maintain
- Each device needs multiple NICs or ports
- Difficult to scale — adding devices requires many new connections

Mesh topologies are used where reliability is critical, such as the internet backbone, military networks, and financial systems.

2.3 The internet

The **internet** is a global network of networks — a worldwide WAN that connects millions of smaller networks together. It is not a single network but rather an interconnection of many networks using standardised protocols (TCP/IP).

Key characteristics of the internet:

- Worldwide scope — connects networks across the globe
- No single owner — decentralised, cooperative structure
- Uses TCP/IP protocols for communication
- Backbone uses mesh topology for reliability
- Provides services: web, email, file transfer, streaming, etc.

2.4 DNS (Domain Name System)

Computers on the internet are identified by **IP addresses** — numerical addresses like 142.250.187.46. However, humans find it easier to remember domain names like "google.com". The **Domain Name System (DNS)** translates human-readable domain names into IP addresses.

How DNS works:

1. You type "www.google.com" into your browser
2. Your computer sends a request to a DNS server
3. The DNS server looks up the domain name in its database
4. The DNS server returns the IP address (e.g., 142.250.187.46)
5. Your computer uses the IP address to connect to the website

DNS servers are distributed around the world, and your request may be passed through several DNS servers before finding the answer. This is called DNS resolution.

2.5 Hosting and the cloud

Hosting means storing website files, applications, or data on servers that are connected to the internet 24/7. When you visit a website, your browser downloads the files from the host server.

Cloud computing refers to accessing computing resources (storage, processing power, software) over the internet instead of using local hardware. Examples include:

- **Cloud storage:** Google Drive, OneDrive, Dropbox — files stored on remote servers
- **Cloud software:** Google Docs, Office 365 — applications run in the browser
- **Cloud hosting:** Websites hosted on scalable cloud infrastructure

Advantages of cloud computing: Access from anywhere, automatic backups, scalability, no need to maintain hardware, often pay-as-you-go pricing.

Disadvantages: Requires internet connection, ongoing costs, data security concerns, dependency on the provider.

> **Key distinction:** DNS translates domain names to IP addresses. Hosting is where website files are stored. The cloud provides remote computing resources accessed via the internet.

Worked Example 5 — Star topology advantages

Question: State two advantages of using a star topology. [2 marks]

Model answer:
1. If one device fails, other devices on the network are not affected **(1)**.
2. It is easy to add new devices to the network by connecting them to the central switch **(1)**.

Exam tip: Other valid advantages: easy to identify faults, good performance, simple to manage. Pick two distinct advantages.

Worked Example 6 — Mesh topology purpose

Question: Explain why mesh topology is used for the internet backbone rather than star topology. [3 marks]

Model answer:
Mesh topology has no single point of failure because devices are connected to multiple other devices **(1)**. If one connection fails, data can be rerouted through alternative paths **(1)**. This high reliability is essential for the internet backbone where downtime would affect millions of users **(1)**.

Exam tip: Focus on reliability and fault tolerance. The internet cannot afford to have a single central switch that could fail.

Worked Example 7 — DNS purpose

Question: Describe the purpose of DNS. [2 marks]

Model answer:
DNS (Domain Name System) translates human-readable domain names (like www.google.com) into IP addresses **(1)**. This allows users to access websites using memorable names rather than having to remember numerical IP addresses **(1)**.

Exam tip: DNS = domain names → IP addresses. Always mention both the "from" (domain name) and "to" (IP address) in your answer.

Worked Example 8 — Cloud computing advantages

Question: A company is considering moving its data to cloud storage. Give two advantages of using cloud storage. [2 marks]

Model answer:
1. Data can be accessed from anywhere with an internet connection **(1)**.
2. The cloud provider handles backups and data security, reducing the company's IT workload **(1)**.

Exam tip: Other valid advantages: scalability, no need to maintain hardware, pay-as-you-go pricing, automatic updates. Apply your answer to the scenario given.

3. Network models and performance

Networks can be organised in different ways depending on their purpose. The two main models are client-server and peer-to-peer. Understanding factors that affect network performance is also essential for exam success.

3.1 Client-server model

In a **client-server** network, one or more dedicated servers provide services to client computers. Clients request services; servers provide them. This is the model used by most business and school networks.

How it works: When you log into your school network, your computer (the client) sends your username and password to the authentication server. The server checks your credentials and either grants or denies access. When you save a file to the network, it goes to a file server. When you print, the request goes to a print server.

Types of servers:

- **File server:** Stores and manages shared files
- **Print server:** Manages print jobs from multiple users
- **Web server:** Hosts websites and responds to browser requests
- **Email server:** Handles sending, receiving, and storing emails
- **Authentication server:** Manages user logins and permissions
- **Database server:** Stores and manages databases

Advantages of client-server:

- Centralised management — easier to secure, backup, and update
- Centralised data storage — users can access files from any client
- Better security — controlled by administrators
- Scalable — easy to add more clients
- Consistent user experience across all clients

Disadvantages of client-server:

- Higher cost — servers are expensive to buy and maintain
- Requires specialist IT staff to manage
- Server failure affects all clients (single point of failure)
- Network traffic can congest around the server

3.2 Peer-to-peer model

In a **peer-to-peer (P2P)** network, all computers are equal — there is no dedicated server. Each computer can share its resources (files, printers) with other computers on the network.

How it works: If Computer A has a file that Computer B wants, B can request it directly from A. There is no central server storing the files. Each computer acts as both a client (requesting resources) and a server (providing resources).

Advantages of peer-to-peer:

- Low cost — no dedicated server hardware needed
- Easy to set up — no specialist IT knowledge required
- No single point of failure — if one computer fails, others continue
- Each user has control over their own shared resources

Disadvantages of peer-to-peer:

- Difficult to manage security — each user is responsible
- No centralised backup — data could be lost if a computer fails
- Performance degrades as the network grows
- Difficult to manage as the number of users increases
- Resources are only available when the sharing computer is on

P2P is suitable for small networks (home networks, small offices) where simplicity and low cost are priorities. It is also used for file-sharing applications where files are distributed across many users.

3.3 Factors affecting network performance

Several factors affect how well a network performs:

- **Bandwidth:** The maximum amount of data that can be transmitted per second (measured in Mbps or Gbps). Higher bandwidth = more data can flow = better performance.
- **Latency:** The delay between sending and receiving data. Lower latency = faster response times. Important for gaming and video calls.
- **Number of users:** More users sharing the same bandwidth means less bandwidth per user. Network can become congested at peak times.
- **Transmission media:** Fibre optic is fastest, copper cable is good for LANs, wireless is convenient but often slower and less reliable.
- **Interference:** Wireless networks can be affected by physical obstacles, other wireless devices, and electronic interference. Wired networks are more reliable.
- **Network hardware:** Outdated switches, routers, or NICs can create bottlenecks. Quality of equipment matters.
- **Distance:** Signals weaken over distance. Longer cables or wireless range = weaker signal = slower speeds.

> **Exam tip:** When asked about improving network performance, consider: upgrading bandwidth, using faster transmission media (e.g., fibre optic), reducing the number of users, minimising interference, or upgrading hardware.

> #### Worked Example 9 — Client-server vs P2P comparison
>
> **Question:** A business with 50 employees is choosing between a client-server and peer-to-peer network. Explain why client-server would be more suitable. [3 marks]

Model answer:
Client-server provides centralised security management, which is important for a business with 50 employees handling potentially sensitive data **(1)**.
Centralised backup on the server protects business-critical data from loss **(1)**.
Client-server scales well and is easier to manage as the business grows **(1)**.

Exam tip: Apply your answer to the scenario. A business with 50 employees is too large for P2P — mention security, backups, and scalability.

Worked Example 10 — Factors affecting performance

Question: State two factors that affect the performance of a network. [2 marks]

Model answer:
1. Bandwidth — the maximum data transfer rate of the network **(1)**.
2. Number of users — more users means bandwidth is shared between more devices **(1)**.

Exam tip: Other valid factors: latency, transmission media, interference, network hardware quality, distance. Give a brief explanation of each factor.

Worked Example 11 — Improving network performance

Question: A school network is running slowly at peak times. Suggest two ways the school could improve network performance. [2 marks]

Model answer:
1. Increase the bandwidth by upgrading the internet connection **(1)**.
2. Add additional wireless access points to reduce congestion and improve coverage **(1)**.

Exam tip: Other valid suggestions: upgrade to faster cables (e.g., Cat6 or fibre), upgrade switches/routers, implement traffic management to prioritise educational content.

Worked Example 12 — Extended comparison (6-mark style)

Question: Compare star topology and mesh topology. Recommend which would be more suitable for a hospital's critical patient monitoring system. [6 marks]

Plan:

- Describe star topology with advantages and disadvantages
- Describe mesh topology with advantages and disadvantages
- Make a recommendation with justification

Model answer:
Star topology: All devices connect to a central switch. Advantages include easy setup, easy fault identification, and if one device fails, others are unaffected. However, star topology has a single point of failure — if the central switch fails, the entire network goes down. This is a significant risk for critical systems.
Mesh topology: Devices connect to multiple other devices, creating redundant paths. Advantages include no single point of failure, high fault tolerance, and data can be rerouted if a connection fails. Disadvantages include higher cost (more cables and connections needed) and complexity in setup and maintenance.
Recommendation: For a hospital's patient monitoring system, mesh topology would be more suitable because patient safety depends on the network being available at all times. The cost of mesh topology is justified because network failure in a hospital could have life-threatening consequences. The redundant paths ensure that even if connections fail, patient data can still be transmitted to medical staff.

Exam tip: For 6-mark questions, cover both options fairly, then give a clear recommendation. Link your recommendation to the specific scenario — here, patient safety justifies the extra cost of mesh.

Worked examples summary

The 12 worked examples above cover the main question types you will encounter on network hardware and topologies. Here is a summary of the key question patterns:

Example	Question type	Key skill
1	Define (2 marks)	LAN vs WAN — geographic scope
2	Describe purpose (2 marks)	Switch — MAC addresses, directed forwarding
3	Explain difference (2 marks)	Router vs switch — different networks vs same network
4	Hardware identification (2 marks)	WAP for wireless connectivity

Example	Question type	Key skill
5	State advantages (2 marks)	Star topology benefits
6	Explain choice (3 marks)	Mesh for reliability — no single point of failure
7	Describe purpose (2 marks)	DNS — domain names to IP addresses
8	Give advantages (2 marks)	Cloud computing benefits
9	Explain suitability (3 marks)	Client-server for larger organisations
10	State factors (2 marks)	Performance factors with brief explanations
11	Suggest improvements (2 marks)	Practical solutions to network problems
12	Compare and recommend (6 marks)	Star vs mesh with justified recommendation

Quick reference

Key terms – exam-ready definitions

Term	Definition (mark scheme style)
LAN	Local Area Network — a network covering a small geographic area such as a building or campus
WAN	Wide Area Network — a network covering a large geographic area, potentially worldwide
NIC	Network Interface Card — hardware that allows a device to connect to a network; has a unique MAC address
Switch	Connects devices within a LAN; uses MAC addresses to forward data to the correct device
Router	Connects different networks together; uses IP addresses to direct data between networks
WAP	Wireless Access Point — allows wireless devices to connect to a wired network
Star topology	Network layout where all devices connect to a central switch; single point of failure
Mesh topology	Network layout where devices connect to multiple other devices; no single point of failure

Term	Definition (mark scheme style)
DNS	Domain Name System — translates domain names (e.g., google.com) into IP addresses
Cloud computing	Accessing computing resources (storage, processing, software) over the internet
Client-server	Network model where dedicated servers provide services to client computers
Peer-to-peer	Network model where all computers are equal and can share resources with each other
Bandwidth	Maximum data transfer rate of a network connection (measured in Mbps/Gbps)
Latency	The delay between sending data and it being received

Hardware comparison

Device	Connects	Uses	Key point
Switch	Devices in same LAN	MAC addresses	Sends to correct device only
Router	Different networks	IP addresses	LAN to internet connection
WAP	Wireless to wired	Radio waves	Creates Wi-Fi hotspot
NIC	Device to network	MAC address	Every device needs one

Topology comparison

Aspect	Star	Mesh
Single point of failure	Yes (central switch)	No
Cost	Lower	Higher
Complexity	Simple	Complex
Reliability	Moderate	Very high
Use case	Most LANs	Critical systems

Typical pitfalls

Pitfall 1 — Confusing switch and router.
Wrong: "A switch connects to the internet."

Right: A **switch** connects devices within a LAN. A **router** connects different networks (e.g., LAN to internet).

Pitfall 2 — Forgetting that star has a single point of failure.
Wrong: "Star topology has no disadvantages."
Right: Star topology has a **single point of failure** — if the central switch fails, the entire network goes down.

Pitfall 3 — Saying DNS "stores websites".
Wrong: "DNS stores web pages."
Right: DNS **translates domain names into IP addresses**. Websites are stored on web servers (hosting).

Pitfall 4 — Confusing LAN and WAN definitions.
Wrong: "A LAN is a wireless network."
Right: LAN = **Local** (small area), WAN = **Wide** (large area). Both can be wired or wireless.

Pitfall 5 — Not knowing what MAC and IP addresses are used for.
Wrong: "Routers use MAC addresses."
Right: Switches use **MAC addresses** (within LAN). Routers use **IP addresses** (between networks).

Pitfall 6 — Thinking P2P is always better because it's cheaper.
Wrong: "P2P is better for businesses because it costs less."
Right: P2P lacks centralised security and backup. For businesses with many users, **client-server is more suitable** despite higher cost.

Pitfall 7 — Confusing bandwidth with speed.
Wrong: "Bandwidth is how fast data travels."
Right: Bandwidth is the **maximum data transfer rate**. Actual speed also depends on latency, congestion, and other factors.

Pitfall 8 — Forgetting what WAP stands for.
Wrong: "WAP is a type of router."
Right: WAP = **Wireless Access Point**. It allows wireless devices to connect to a wired network. It is not a router.

Pitfall 9 — Not applying answers to the scenario.
Wrong: "Mesh is good because it has no single point of failure."

Right: "Mesh is suitable for the hospital because **patient monitoring cannot afford network downtime**, so the redundant connections are worth the extra cost."

Pitfall 10 — One-sided comparisons.
Wrong: Only giving advantages of one option.
Right: When comparing, give **advantages and disadvantages of both** options, then make a recommendation with justification.

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